

APPOINTMENTS

Sep 2026– **Postdoctoral Researcher, MIT**
Jun 2026– **Visiting Scholar, University of California, Berkeley** (Host: Umesh Vazirani)

EDUCATION

2020–2026 **Ph.D. in Computer Science, University of Chicago** (Advisor: Bill Fefferman)
– Recipient of the **IBM PhD Fellowship (2026)**.
2018–2020 **M.Math. in Computer Science, University of Waterloo** (Advisor: John Watrous)
2014–2018 **B.E. in Electronics Engineering, Jadavpur University, India**

PUBLICATIONS

Google Scholar • Newest first; author order is typically alphabetical in theoretical computer science.

1. S. Martiel, J.-U. Chung, A. Seif, **S. Ghosh**, I. Hincks, A. Deshpande, B. Fefferman, J. M. Gambetta & A. Javadi-Abhari. Sampling hard circuits with verifiably high fidelity. *arXiv* (2026).
2. G. Bentsen, B. Fefferman, **S. Ghosh**, M. J. Gullans & Y. Liu. Sample-efficient benchmarking of shallow all-to-all random quantum circuits. *arXiv* (2026).
3. A. A. Mele, A. Angrisani, **S. Ghosh**, S. Khatri, J. Eisert, D. S. França & Y. Quek. Noise-induced shallow circuits and the absence of barren plateaus. *Nature Physics* (2026).
– Accepted as a talk at **TQC 2024**.
4. P. Niroula, M. Liu, S. Omanakuttan, D. Amaro, S. Chakrabarti, **S. Ghosh**, Z. He, Y. Jin, *et al.* Digital signatures with classical shadows on near-term quantum computers. *arXiv* (2026).
5. A. Bansal, I. George, **S. Ghosh**, J. Sikora & A. Zheng. Online learning of a panoply of quantum objects. *Quantum Machine Intelligence* (2025).
6. S. Chakraborty, S. Choi, **S. Ghosh** & T. Giurgică-Tiron. Fast computational deep thermalization. *Physical Review Letters* (2025).
7. **S. Ghosh**, A. Mirani, Y. Quek & M. Xu. Design boosters: from constant-time quantum chaos to ∞ -designs and beyond. *arXiv* (2025).
8. **S. Ghosh**, D. Hangleiter & J. Helsen. Random regular graph states are complex at almost any depth. *PRX Quantum* (2025).
9. S.-u. Lee, **S. Ghosh**, C. Oh, K. Noh, B. Fefferman & L. Jiang. Classical simulation of noisy random circuits from exponential decay of correlation. *arXiv* (2025).
– Accepted as a talk at **TQC 2026**.
10. A. Deshpande, B. Fefferman, **S. Ghosh**, M. Gullans & D. Hangleiter. Peaked quantum advantage using error correction. *arXiv* (2025).
– Invited workshop talk at **FOCS 2025**.
11. J. T. Iosue, Y.-X. Wang, I. Datta, **S. Ghosh**, C. Oh, B. Fefferman & A. V. Gorshkov. Higher moment theory and learnability of bosonic states. *arXiv* (2025).
– Accepted as a talk at **TQC 2026**.
12. **S. Ghosh**, S. Subramanian & W. Zhan. Unconditional Pseudorandomness against Shallow Quantum Circuits. *arXiv* (2025).
– Accepted as a talk at **ITCS 2026**.
13. B. Fefferman, **S. Ghosh**, M. Sinha & H. Yuen. The Hardness of Learning Quantum Circuits and its Cryptographic Applications. *arXiv* (2025).
– Accepted as a talk at **QCTIP 2025**.
– Accepted as a talk at **ITCS 2026**.
14. D. Belkin, J. Allen, **S. Ghosh**, C. Kang, S. Lin, J. Sud, F. T. Chong, B. Fefferman & B. K. Clark. Approximate t-Designs in Generic Circuit Architectures. *PRX Quantum* (2024).
15. G. Bentsen, B. Fefferman, **S. Ghosh**, M. J. Gullans & Y. Liu. On the complexity of sampling from shallow Brownian circuits. *arXiv* (2024).
16. R. Jozsa, **S. Ghosh** & S. Strelchuk. IQP computations with intermediate measurements. *arXiv* (2024).
17. B. Fefferman, **S. Ghosh** & W. Zhan. Anti-Concentration for the Unitary Haar Measure and Applications to Random Quantum Circuits. *arXiv* (2024).
– Accepted as a talk at **QIP 2025**.
– Accepted as a talk at **ITCS 2026**.
18. B. Fefferman, **S. Ghosh**, M. Gullans, K. Kuroiwa & K. Sharma. Effect of Nonunitary Noise on Random-Circuit Sampling. *PRX Quantum* (2024).
– Accepted as a talk at **QIP 2024**.
19. A. Bouland, B. Fefferman, **S. Ghosh**, T. Metger, U. Vazirani, C. Zhang & Z. Zhou. Public-Key Pseudoentanglement and the Hardness of Learning Ground State Entanglement Structure. Accepted in proceedings of *Computational Complexity Conference (CCC)* (2024).
– Accepted as a talk at **QIP 2024**.
20. A. Gu, L. Leone, **S. Ghosh**, J. Eisert, S. F. Yelin & Y. Quek. Pseudomagic Quantum States. *Physical Review Letters* (2024).
– Editor's choice at PRL.
21. S. Aaronson, A. Bouland, B. Fefferman, **S. Ghosh**, U. Vazirani, C. Zhang & Z. Zhou. Quantum Pseudoentanglement. Accepted in proceedings of *Innovations in Theoretical Computer Science (ITCS)* (2024).
– Accepted as a talk at **QIP 2023**.
– Invited to a **Quantum Colloquium** at the Simons Institute.
22. **S. Ghosh**, A. Deshpande, D. Hangleiter, A. V. Gorshkov & B. Fefferman. Complexity Phase Transitions Generated by Entanglement. *Physical Review Letters* (2023).
23. **S. Ghosh** & J. Watrous. Complexity Limitations on One-turn Quantum Refereed Games. *Theory of Computing Systems* (2023).

SELECTED INVITED TALKS

Peaked quantum advantage using error correction

- July 2026 Quantum Summer Cluster Final Workshop, Simons Institute
- December 2025 CS Theory seminar, UT Austin
- October 2025 CS Theory seminar, University of Illinois Urbana–Champaign
- October 2025 Quantum seminar, MIT

Quantum pseudoentanglement

- August 2025 Tutorial, Kavli Institute for Theoretical Physics, UC Santa Barbara
- August 2025 Many-Body Quantum Magic Workshop, IQuS, University of Washington
- September 2024 Computer Science Theory seminar, Columbia University
- July 2024 Physics seminar, University of Tokyo
- March 2024 Quantum Reunion Workshop, Simons Institute
- September 2023 Colloquium, Tata Institute of Fundamental Research
- August 2023 Quantum seminar, MIT
- July 2023 DAMTP seminar, University of Cambridge
- March 2023 Quantum colloquium, University of Maryland
- February 2023 CS Theory seminar, UT Austin

Effect of non-unital noise on random-circuit sampling

- September 2024 Qiskit seminar, IBM

Learning and cryptography with random circuits

- August 2025 CS Theory seminar, University of Washington
- July 2025 Quantum Research seminar, JP Morgan
- June 2025 Seminar, Google Quantum
- March 2025 Quantum colloquium, CWI Amsterdam
- December 2024 BIRS workshop, Chennai Mathematical Institute

SELECTED MEDIA COVERAGE

- 2026 **The Wall Street Journal**, IBM Claims New Era of "Quantum Advantage"
- 2026 **UChicago News**, IBM, UChicago demonstrate "quantum advantage"
- 2026 **Live Science**, IBM scientists claim they've achieved "quantum advantage"
- 2025 **New Scientist**, Quantum computers could protect our data from quantum computers
- 2025 **Shtetl Optimized**, Quantum computing: too much to handle!
- 2024 **Quanta Magazine**, The Quest to Quantify Quantumness
- 2023 **Quantum Insider**, Entanglement and computational hardness in quantum systems

ACADEMIC SERVICE

Reviewer for QIP, TQC, STOC, FOCS, *Physical Review Letters*, *Nature Physics*, PRX, PRX Quantum, and *Quantum*.